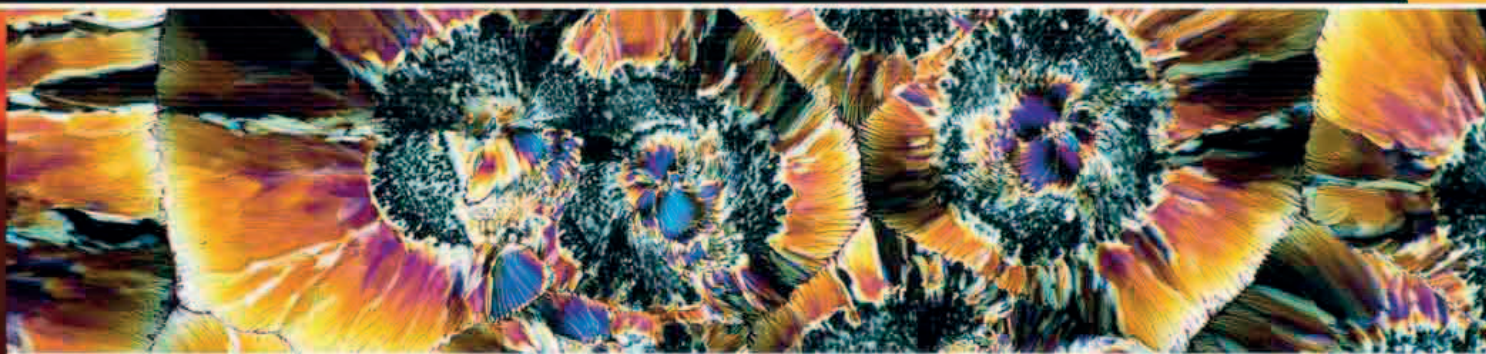


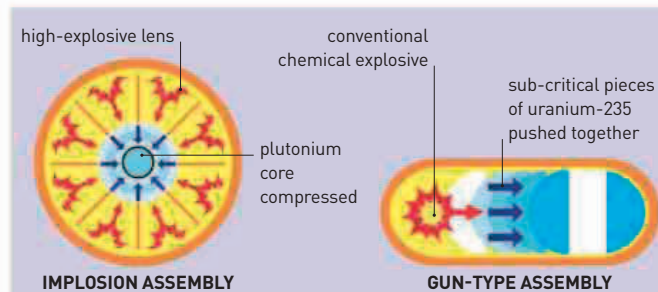


This polarized light micrograph shows crystals of the vitamin folic acid (folacin), which is needed during pregnancy for normal development of the baby in the womb.

AS WORLD WAR II ENTERED ITS SIXTH YEAR, so did the **Manhattan Project**—the Allies' atomic bomb research project (see 1939). The idea of an artificial nuclear chain reaction had been developed by Hungarian-born physicist **Léó Szilárd** (1898–1964), but it did not become a reality until 1942. Nuclear reactions are processes that change the nuclei of atoms, either through **fission** (splitting them) or **fusion** (combining

The world's first atomic bomb

The first bomb was exploded by the US Army in the early hours of July 16, 1945, at Alamogordo, in the New Mexico desert. It had the energy equivalent of 20,000 tons of TNT.



BUILDING ATOMIC BOMBS

Two methods have been used. In the implosion method (used for the Trinity Test and the Nagasaki bomb), explosives compress a central core of fissile material. In the gun-type assembly (used on Hiroshima), elements were pushed together. The naturally emitted neutrons strike neighboring atoms, causing a chain reaction of fission events (see 1938), which releases huge amounts of energy.

“WHEN YOU SEE SOMETHING THAT IS **TECHNICALLY SWEET**, YOU GO AHEAD AND DO IT. THAT IS THE WAY IT WAS WITH THE **ATOMIC BOMB**.”

J. Robert Oppenheimer, American physicist, testifying in court, 1954

them). Both release energy, but only **fission could be achieved artificially**. Fission happens naturally when radioactive elements decay, but it can be **induced by bombarding elements with neutrons** (see 1938). If elements capable of sustaining a nuclear fission chain reaction (fissile material), such as uranium or plutonium, are condensed to a critical mass,

the neutrons emitted by the radioactivity trigger instantaneous decay and release massive amounts of energy. On July 16, the US Army exploded the **Manhattan Project's first atomic bomb** in the New Mexico desert—the Trinity Test—witnessed by fewer than 300 people. Three weeks later, two atomic bomb attacks on Japan (on **Hiroshima**, and **Nagasaki**), helped bring World War II to an end, but hundreds of thousands of lives were lost.

Scientists had already developed X-ray crystallography, a technique that could be used to work out positions of a crystal's atoms (see 1912), and it proved especially useful for studying structures of complex biological molecules. By July, British chemist **Dorothy Hodgkin** had helped resolve the complex **structures of both penicillin and cholesterol**.

In August, chemical company American Cyanamid announced that it had made **folic acid**—the vitamin needed for healthy growth in the developing fetus.



DOROTHY HODGKIN
(1910–94)

Born in Egypt, Dorothy Hodgkin studied chemistry at Oxford University, UK, and earned a PhD in the study of biological substances called sterols. She studied the three-dimensional structure of complex molecules such as cholesterol, penicillin, and insulin. She was awarded the 1964 Nobel Prize for her research into vitamin B12.

British writer **Arthur C. Clarke** (1917–2008) was looking to the future. Among his many predictions was the idea that **geostationary satellites**, (satellites that would sit at a fixed point in space relative to Earth's position and rotation period) could be **used for telecommunications**. This became a reality less than 20 years later.

May Dorothy Hodgkin confirms structure of penicillin with a three-dimensional map

July 23 Dorothy Hodgkin and British scientist **Harry Cartlidge** describe the crystal structure of a form of cholesterol

August 9 A second B-29 named Bockscar drops a **gun-type assembly atomic bomb** (codenamed Fat Man) on Nagasaki

August 6 A USAAF Boeing B-29 Superfortress—the Enola Gay—drops an **implosion assembly atomic bomb** (codenamed Little Boy) on Hiroshima

August 31 American Cyanamid announces that it has extracted and **purified crystals of folic acid**

September 13 American chemists **Burris Cunningham** and **Louis Werner** isolate the element **americium**

October Arthur C. Clarke describes the idea of a **geosynchronous communications satellite**